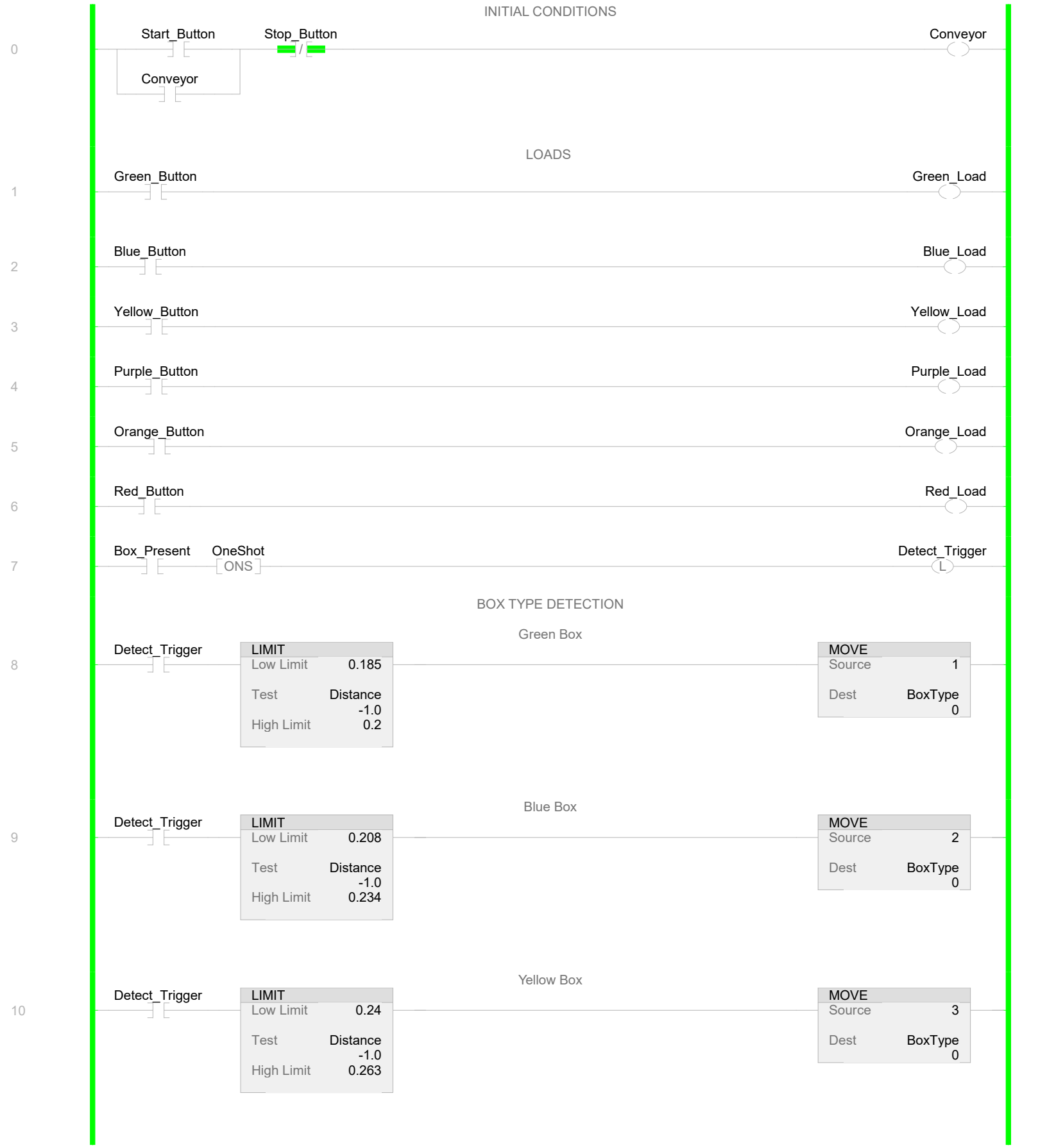

INITIAL CONDITIONS
Start_Button
Stop_Button
Conveyor
Conveyor
LOADS
Green_Button
Green_Load
Blue_Button
Blue_Load
Yellow_Button
Yellow_Load
Purple_Button
Purple_Load
Orange_Button
Orange_Load
Red_Button
Red_Load
Box_Present
OneShot
ONS
Detect_Trigger
L
BOX TYPE DETECTION
Green Box
Detect_Trigger
LIMIT
Low Limit 0.185
Test Distance -1.0
High Limit 0.2
MOVE
Source 1
Dest BoxType 0
Blue Box
Detect_Trigger
LIMIT
Low Limit 0.208
Test Distance -1.0
High Limit 0.234
MOVE
Source 2
Dest BoxType 0
Yellow Box
Detect_Trigger
LIMIT
Low Limit 0.24
Test Distance -1.0
High Limit 0.263
MOVE
Source 3
Dest BoxType 0

Purple Box

11 Detect_Trigger

LIMIT
Low Limit 0.264
Test Distance -1.0
High Limit 0.272

MOVE
Source 4
Dest BoxType 0

Orange Box

12 Detect_Trigger

LIMIT
Low Limit 0.273
Test Distance -1.0
High Limit 0.281

MOVE
Source 5
Dest BoxType 0

Red Box

13 Detect_Trigger

GT
Source A Distance -1.0
Source B 0.281

LIMIT
Low Limit 0.281
Test Distance -1.0
High Limit 0.3

MOVE
Source 6
Dest BoxType 0

POSITION DETECTION

Position Green

14 CollisionSensor_Green

EQ
Source A BoxType 0
Source B 1

Stopper_Green_Cmd (L)

15 Stopper_Green_Cmd /

Stopper_Green

Position Blue

16 CollisionSensor_Blue

EQ
Source A BoxType 0
Source B 2

Stopper_Blue_Cmd (L)

17 Stopper_Blue_Cmd

Stopper_Blue

Position Yellow

18 CollisionSensor_Yellow

EQ
Source A BoxType 0
Source B 3

Stopper_Yellow_Cmd (L)

19 Stopper_Yellow_Cmd /

Stopper_Yellow

Position Purple

20 CollisionSensor_Purple — EQ (Source A: BoxType, 0; Source B: 4) — Stopper_Purple_Cmd (L)

21 Stopper_Purple_Cmd — Stopper_Purple

Position Orange

22 CollisionSensor_Orange — EQ (Source A: BoxType, 0; Source B: 5) — Stopper_Orange_Cmd (L)

23 Stopper_Orange_Cmd (/) — Stopper_Orange

Position Red

24 CollisionSensor_Red — EQ (Source A: BoxType, 0; Source B: 6) — Stopper_Red_Cmd (L)

25 Stopper_Red_Cmd (/) — Stopper_Red

PLUNGER ACTIVATION

Plunger Green

26 Stopper_Green_Cmd — TON (Timer: Timer_Green; Preset: 300; Accum: 0) — EN, DN

27 Timer_Green.DN — Plunger_Green (L)

Plunger Blue

28 Stopper_Blue_Cmd — TON (Timer: Timer_Blue; Preset: 300; Accum: 0) — EN, DN

29 Timer_Blue.DN — Plunger_Blue (L)

Plunger Yellow

30 Stopper_Yellow_Cmd — TON (Timer: Timer_Yellow; Preset: 300; Accum: 0) — EN, DN

31 Timer_Yellow.DN — Plunger_Yellow (L)

Plunger Purple

32 Stopper_Purple_Cmd

TON
Timer Timer_Purple
Preset 300
Accum 0
EN
DN

33 Timer_Purple.DN

Plunger_Purple
L

Plunger Orange

34 Stopper_Orange_Cmd

TON
Timer Timer_Orange
Preset 300
Accum 0
EN
DN

35 Timer_Orange.DN

Plunger_Orange
L

Plunger Red

36 Stopper_Red_Cmd

TON
Timer Timer_Red
Preset 300
Accum 0
EN
DN

37 Timer_Red.DN

Plunger_Red
L

Plunger Trajectory End

38 Plunger_Green
Plunger_Blue
Plunger_Yellow
Plunger_Purple
Plunger_Orange
Plunger_Red

TON
Timer Plunger_Timer
Preset 500
Accum 0
EN
DN

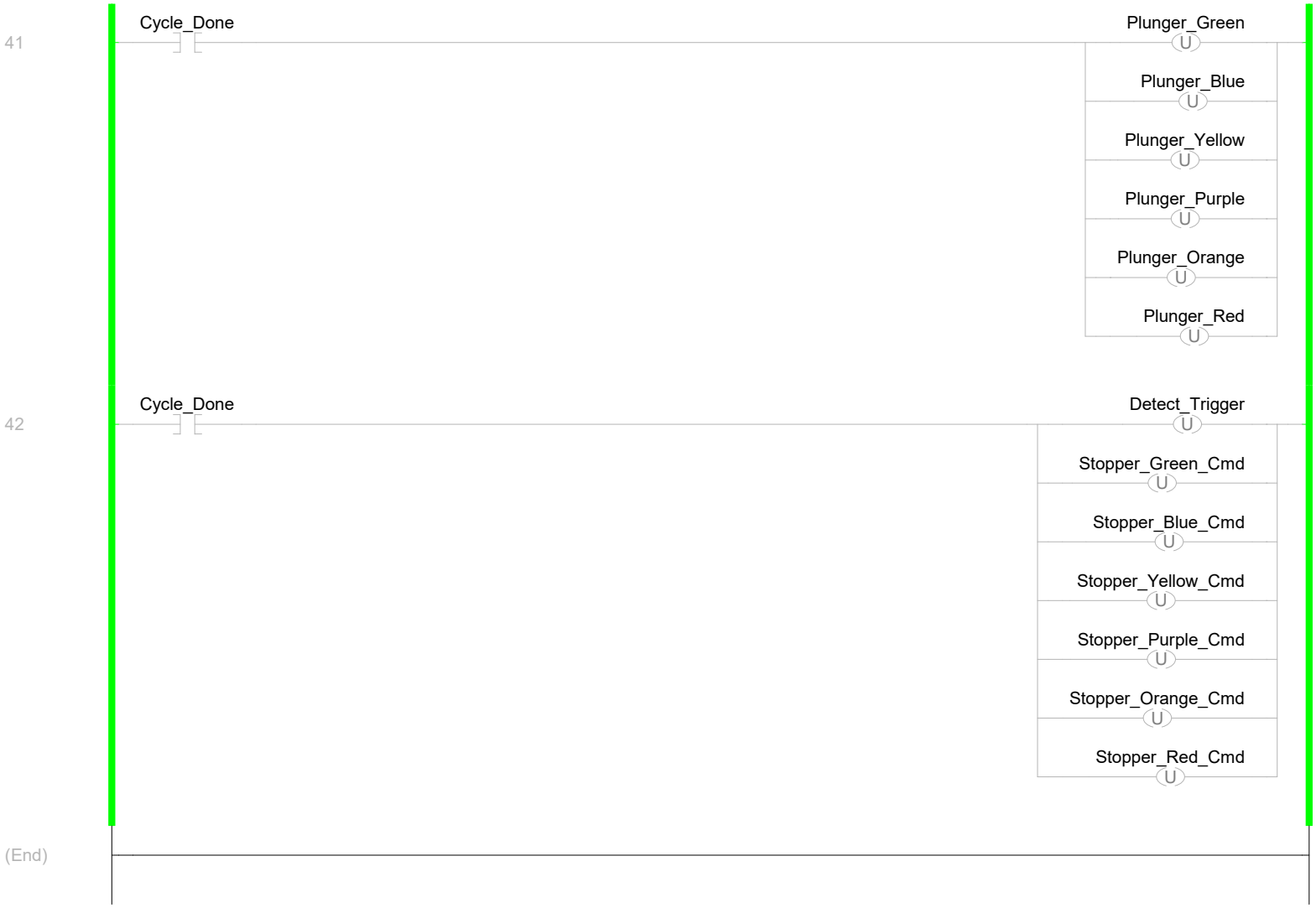
RESTART

39 Plunger_Timer.DN

Cycle_Done

40 Cycle_Done

MOVE
Source 0
Dest BoxType
0


41
Cycle_Done
Plunger_Green
U
Plunger_Blue
U
Plunger_Yellow
U
Plunger_Purple
U
Plunger_Orange
U
Plunger_Red
U
42
Cycle_Done
Detect_Trigger
U
Stopper_Green_Cmd
U
Stopper_Blue_Cmd
U
Stopper_Yellow_Cmd
U
Stopper_Purple_Cmd
U
Stopper_Orange_Cmd
U
Stopper_Red_Cmd
U
(End)